

6.1 Deformation: Stress & Strain

Question Paper

Course	CIE A Level Physics
Section	6. Deformation of Solids
Topic	6.1 Deformation: Stress & Strain
Difficulty	Easy

Time allowed: 20

Score: /10

Percentage: /100

Question 1

A student wanted to measure the Young modulus of a wire, to do this they needed to take a number of measurements.

In which row could the measurement **not** be made **directly** with the listed apparatus?

	Measurement	Apparatus
A	extension of wire	vernier scale
B	area of cross-section of wire	micrometer screw gauge
C	original length of wire	metre rule
D	mass of load applied to a wire	electronic balance

[1 mark]

Question 2

What is the unit of the Young modulus?

- A. N m^{-2}
- B. N m
- C. N m^{-1}
- D. N m^2

[1 mark]

Question 3

Which of the following is the definition of the ultimate tensile stress of a material?

- A. the maximum inter-atomic force before the atomic bonds of the material break
- B. the maximum force that can be applied to a bar of the material before it bends
- C. the maximum tensile force in a wire of the material before it breaks
- D. the maximum stretching force per unit cross-sectional area before the material breaks

[1 mark]

Question 4

A material is extended elastically within the limit of proportionality. What is equal to the Young modulus of the material?

- A. gradient of the stress-strain graph
- B. area under the force-extension graph
- C. area under the stress-strain graph
- D. gradient of the force-extension graph

[1 mark]**Question 5**

An experiment was carried out to determine the Young modulus of steel using a steel wire.

The uncertainties in the measurements are given below.

Measurement	Uncertainty
load on wire	$\pm 2\%$
length of wire	$\pm 0.2\%$
diameter of wire	$\pm 1.5\%$
extension	$\pm 1\%$

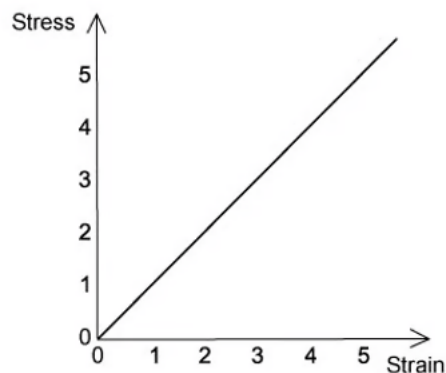
What is the percentage uncertainty in the Young modulus?

- A. 6.2%
- B. 4.7%
- C. 1.8%
- D. 1.3%

[1 mark]

Question 6

A glass rod was put under stress and strain until it broke. The graph of this is shown below.



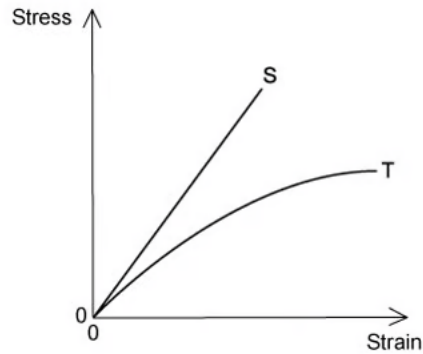
Which of the following statements is correct about the glass rod?

- A. the glass shows plastic deformation
- B. when the cross-sectional area of the rod is doubled, the ultimate tensile stress of the rod is halved
- C. Hooke's law is obeyed for all values of stress up to the breaking point
- D. the glass is ductile

[1 mark]

Question 7

Two wires S and T, with the same initial dimensions, were put under stress and strain, the graph shows the results up to their breaking points.



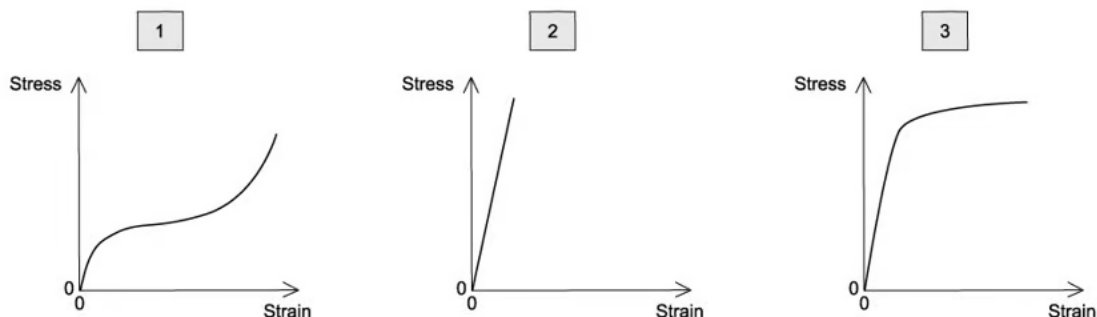
Which statement is **not** correct?

- A. material S has a larger ultimate tensile stress
- B. material S extends elastically
- C. material S extends more than material T when loaded with the same force
- D. material S is brittle

[1 mark]

Question 8

The graphs below show the stress-strain graphs of three materials. The graphs do not have the same scales.



The three materials are copper, rubber and glass.

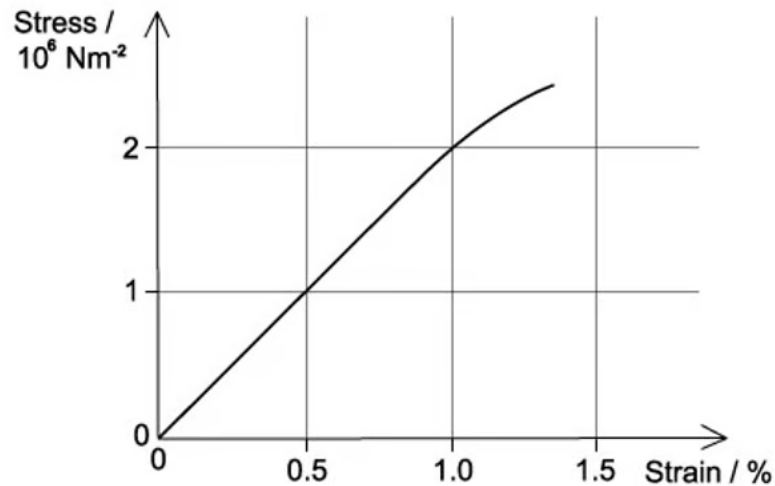
Which materials are represented by the graphs?

	1	2	3
A	rubber	copper	glass
B	rubber	glass	copper
C	glass	copper	rubber
D	copper	glass	rubber

[1 mark]

Question 9

The stress-strain graph for bone is shown below.



What is the Young modulus of bone?

- A. $2 \times 10^6 \text{ Nm}^{-2}$
- B. $2 \times 10^8 \text{ Nm}^{-2}$
- C. $1 \times 10^6 \text{ Nm}^{-2}$
- D. $1 \times 10^8 \text{ Nm}^{-2}$

[1 mark]

Question 10

A student wanted to determine the spring constant of a spring. They did this by using different loads to extend the spring by varying amounts.

To find the spring constant, which quantities are required?

- A. the elastic limit, extensions and the length of the spring
- B. the loads and the extensions of the spring
- C. the elastic limit and the loads
- D. the loads and the length of the spring

[1 mark]

Model Answers: Easy

1

The correct answer is **B** because:

- The area of the cross-section of the wire = πr^2
- A micrometer can measure the diameter of the wire with a resolution of 0.001 mm
- Therefore, a micrometer cannot directly measure the cross-sectional area of the wire

A is incorrect because the vernier scale will measure the extension of the wire

C is incorrect because the length of wire could be measured directly by a metre rule

D is incorrect because an electronic balance will measure the mass of the load directly

2

The correct answer is **A** because:

- The Young modulus is given by $E = \frac{\text{stress}}{\text{strain}} = \frac{\sigma}{\epsilon}$
 - Stress is equal to $\sigma = \frac{\text{force}}{\text{area}} = \frac{F}{A} = [N][m^2]^{-1} = N m^{-2}$
 - Strain is equal to $\epsilon = \frac{\text{extension}}{\text{original length}} = \frac{\Delta L}{L} = [m][m]^{-1} = 1$
- Strain is dimensionless, so, the units for the Young modulus are $N m^{-2}$

3

The correct answer is **D** because:

- The ultimate tensile stress (UTS) is defined as the maximum stress a material can withstand while being stretched or pulled before breaking

- UTS is measured as force per unit area and has the units Pa (N m^{-2})

A is incorrect because inter-atomic forces, also known as intermolecular forces, are the forces that act between the molecules

B & C are incorrect because the force can be applied to any shape of the material and not just a bar

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The correct answer is **A** because:

- The Young modulus is given by $E = \frac{\text{stress}}{\text{strain}} = \frac{\sigma}{\epsilon}$
 - Stress is equal to $\sigma = \frac{\text{force}}{\text{area}} = \frac{F}{A}$
 - Strain is equal to $\epsilon = \frac{\text{extension}}{\text{original length}} = \frac{\Delta L}{L}$
- The Young modulus is equal to the gradient of the stress-strain graph

B is incorrect because the area under a force-extension graph is the energy stored in the spring

C is incorrect because the area under a stress-strain graph is the toughness

D is incorrect because the gradient of a force-extension graph is the spring constant, k

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The correct answer is **A** because:

- The equation for Young Modulus is:
 - $\text{Young Modulus} = \frac{\text{Force}(F)}{\text{Area}(A)} \times \frac{\text{length}(l)}{\text{extension}(x)} = \frac{\text{load} \times \text{length}}{\frac{\pi}{4} \times \text{diameter}^2 \times \text{extension}}$
- To find the percentage uncertainty in Young modulus, we must add the uncertainties in the measurements
- We can ignore the $\frac{\pi}{4}$ since it is a constant with no units, and it is important to

multiply the uncertainty in the diameter by 2 since the diameter is squared

◦ % uncertainty = $2 + 0.2 + (2 \times 1.5) + 1 = 6.2\%$

6

The correct answer is **C** because:

- Hooke's law states that the strain in a solid is proportional to the applied stress within the elastic limit
- The graph is a straight line going through the origin, showing proportionality
- Therefore, Hooke's law is obeyed for all values of stress up to the breaking point

A is incorrect because plastic deformation occurs after the limit of proportionality and is after the elastic limit, glass is brittle and does not show plastic deformation

B is incorrect because the ultimate tensile strength is the maximum stress a material can withstand before breaking. This is usually a constant value, if the cross-sectional area was doubled the ultimate tensile strength would be twice

D is incorrect because glass is brittle and does not show plastic deformation as a ductile material would

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The correct answer is **C** because:

- The Young modulus is given by $E = \frac{\text{stress}}{\text{strain}} = \frac{\sigma}{\epsilon}$
 - Stress is equal to $\sigma = \frac{\text{force}}{\text{area}} = \frac{F}{A}$
 - Strain is equal to $\epsilon = \frac{\text{extension}}{\text{original length}} = \frac{\Delta L}{L}$
- The strain is determined by the extension of the material and T has a larger strain than S

A is incorrect because the ultimate tensile strength is the maximum stress a material can withstand before breaking; this is higher for S than T

B is incorrect because material S shows proportionality and extends elastically

D is incorrect because material S shows no plastic deformation and is brittle

8

The correct answer is **B** because:

- Material 1 shows an unusual shape stress-strain graph associated with a polymer, such as rubber
 - Rubber deforms elastically up to a certain point as the polymers chains, which were previously curled up, uncurl
 - After its yield point is reached it will no longer return to its original shape as it undergoes hysteresis
- Material 2 shows a brittle material that shows no plastic deformation, such as glass
 - Glass is described as brittle as it does not extend very much and when subjected to stress, it breaks with little elastic deformation
- Material 3 shows a ductile material that shows plastic deformation, such as a metal wire
 - A metal wire tends to be described as ductile as it will experience a large amount of extension when subjected to stress, especially beyond the elastic limit

9

The correct answer is **B** because:

- The Young modulus is given by $E = \frac{\text{stress}}{\text{strain}} = \frac{\sigma}{\epsilon}$
 - Stress is equal to $\sigma = \frac{\text{force}}{\text{area}} = \frac{F}{A}$
 - Strain is equal to $\epsilon = \frac{\text{extension}}{\text{original length}} = \frac{\Delta L}{L}$
- From the graph:
 - Stress = $2 \times 10^6 \text{ N m}^{-2}$
 - Strain = $1\% = \frac{1}{100} = 0.01$

- Strain is equal to $\epsilon = \frac{\text{extension}}{\text{original length}} = \frac{\Delta L}{L}$
- From the graph:
 - Stress = $2 \times 10^6 \text{ N m}^{-2}$
 - Strain = $1\% = \frac{1}{100} = 0.01$
- Young modulus = $\frac{2 \times 10^6}{0.01} = 2 \times 10^8 \text{ N m}^{-2}$

Exam Tip: a common error students make is that they tend to forget to check the units on the graph's axes. It would be an easy mistake to make to take the percentage strain from the graph and not convert it back to the ratio extension/original length.

10

The correct answer is **B** because:

- Hooke's law states that the strain in a solid is proportional to the applied stress within the elastic limit
- Hooke's law states that $F = kx$, where k is the spring constant, F is the force and e is the extension
 - So, to calculate the spring constant $k = \frac{F}{x}$
- To do this, the force (load) and the extension of the spring are required

A & D are incorrect because the elastic limit is the point that the material will no longer obey Hooke's law, it is calculated using the extension not the length of the material

C is incorrect because the elastic limit is the point that the material will no longer obey Hooke's law